

LDMX HCal prototype data analysis

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THESIS SUBMITTED FOR DEGREE OF MASTER OF SCIENCE

PROJECT DURATION: 9 MONTHS

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June 20, 2025

Dedication

Dedicated to my grandfather, who was the first to cultivate my curiosity for the world and who never stopped believing in me. I miss you.

Nagypapinak, aki elsőként kezdted ápolni bennem a kíváncsiságot a világ iránt és soha nem szüntél meg hinni bennem. Hiányzol.

Gréti

Abstract

The Light Dark Matter eXperiment (LDMX) is a planned missing momentum and energy experiment at SLAC, the purpose of which is to produce dark matter through production at an accelerator. A key theoretical process making this possible is called “dark bremsstrahlung,” where an electron scattering off a nucleus produces a dark photon instead of a Standard Model photon. LDMX aims to detect dark matter production through the invisible signature: an electron with missing energy and a significant transverse momentum change, with no other particles present. Dark bremsstrahlung has multiple background processes that could result in the production of neutral hadrons, making a hadronic calorimeter (HCal) necessary for vetoing these events. However, detection through visible signatures, such as the dark photon decaying back into Standard Model particles, is also possible. As this decay has a high chance of occurring in the HCal, identifying the decay products and reconstructing their energies would be important to be able to characterize the dark photon. A test beam was conducted in 2022 at CERN with a prototype version of the HCal. The electron energy response and resolution, as well as shower shapes, were determined using data taken at the test beam and found to be consistent with expectations.

Acknowledgments

First and foremost, I would like to thank my supervisors Hannah and Erik for all the wonderful support and wisdom they have provided me with during the past year. I would also like to thank the LunDMX group and the participants of the ATLAS-LDMX fika meetings for the invaluable discussions. I would like to thank my office mates Axel and Xintong for being good company and great conversation partners. I would like to thank Tommy for providing chicken content and helping me become a better communicator and educator (and Tommy's chickens, for existing). I would like to thank my friends and family for believing in me and providing constant support. And also, my bunny PomPom for being adorable and cheering me up when needed.

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List of acronyms

HCal	H adronic C alorimeter
ECal	E lectromagnetic C alorimeter
LDMX	L ight D ark M atter eX periment
DM	D ark M atter
SM	S tandard M odel
SUSY	S upersymmetry
MC	M onte C arlo
ADC	A nalog- D igital C onverter
WIMP	W eakly I nteracting M assive P article
LDM	L ight D ark M atter
TDC	T ime-to- D igital C onverter
TOT	T ime- O ver- T hreshold
TOA	T ime- O f- A rrival
LESA	L inac to E nd S tation A
ESA	E nd S tation A
LHC	L arge H adron C ollider
EM	E lectrom m agnetic
MIP	M inimum I onizing P article

1 Introduction

The Light Dark Matter eXperiment (LDMX) is a planned missing momentum and energy experiment at SLAC. Its purpose is to discover dark matter (DM) through production at an accelerator. A key theoretical process making this possible is called “dark bremsstrahlung”. Similar to Standard Model (SM) bremsstrahlung, an electron is scattered off of a nucleus. However, instead of a SM photon, it produces a dark photon, which serves as a mediator between SM particles and DM. By definition, even if LDMX is successful in producing DM, DM itself would be invisible to any detector that could be built. While detection is possible through, for example, the dark photon decaying back into SM particles, such events would be doubly suppressed due to the supposed small coupling between the dark photon and SM particles. LDMX aims to instead detect DM through an invisible signature - if the electron loses energy and has a significant change in transverse momentum, but no other particle is present, then DM was produced.

The LDMX detector concept is built around this signal. Electrons will be shot at a thin tungsten target - that will provide the nuclei to decelerate our electron. Trackers will be located before and after the target, allowing for the measurement of the electron’s momenta in both stages. An electromagnetic calorimeter (ECal) will be placed after the second tracker to “catch” the electron and measure its energy, thereby making the missing energy calculable. It can also capture any other electromagnetically-interacting particles resulting from background processes, e.g. a photon coming from SM bremsstrahlung.

Nevertheless, other electro-nuclear and photonuclear background processes may result in the production of neutral particles, making a hadronic calorimeter (HCal) necessary. Although the HCal’s main purpose will be to serve as a veto device to remove these events, it might also be used to detect visible DM signatures. Since the dark photon has a chance to decay back into SM particles instead of DM, that decay may happen in the HCal. In that case, identifying the decay products and reconstructing their energies would be important to characterize the dark photon.

A test beam was conducted in 2022 at CERN with a prototype version of the HCal. Electrons, pions, and muons were shot directly into the prototype at different energies (300 MeV to 4 GeV) allowing us - among other things - to study the signatures they make in the HCal. This thesis aims to explore the potential of visible DM signatures in the LDMX HCal by analyzing this data. Specifically, electron energy resolution and electron shower shapes were determined with this goal in mind.

2 The problem of Dark Matter

Humans have been occupied with the idea of invisible matter since time immemorial. Too small, too distant, or just too *intrinsically invisible*, it moved the fantasy of our ancestors all the same [1]. Dark matter belongs in this category. Today, it is usually defined as a type of hypothetical matter that does not interact with photons, but does interact through gravity. While often when one talks about DM, they mean *particle* DM, this has not always been the case [1].

The first inklings of the DM problem popped up in the late 19th century. Newton’s law of universal gravitation has already allowed astronomers to predict the existence of “invisible” objects through their gravitational pull, such as the planet Uranus. It was only natural that astronomers began thinking about just *how many* of these objects could possibly be out there, and what they might be. Since stars were not distributed evenly on the night sky, they were wondering what was in the “unfilled” spaces. Was it simply empty, or was there a “dark cloud”, obscuring their vision [1]?

Lord Kelvin was the first to attempt to make calculations to estimate the amount of stars in the Milky Way. He questioned what the upper limit for matter in a sphere with a radius of 3.09×10^{16} km would be. He concluded, to agree with the observed velocity dispersion of the stars, the maximum mass would have to be equivalent to the mass of a billion suns, effectively calculating an upper boundary for density. However, this was many more stars than could be observed in his day, leading to the conclusion that many of the supposed billion stars may be dark bodies, with light too faint to be detectable [1][2][3].

Over the following decades, many astronomers made similar predictions and models, attempting to estimate the total mass of DM in the Milky Way. Öpik, Kapteyn, Oort, Jeans, and Lindbland all made such calculations, all coming to the same conclusion: the amount of DM, while considerable, is still less than the total mass of visible stars [1].

A dissent from this point of view came from Knut Lundmark in 1930. He calculated the mass-to-light ratio of five galaxies, using their luminosities and estimated masses from spectroscopic observations. The results were much larger than anticipated, which pointed to more DM than previously thought. A similar calculation was done on the Coma Cluster (pictured in Fig. 1) by Zwicky. The fact that some galaxies in the cluster had much higher velocity dispersions than expected was already known through the work of Hubble and Humason. Zwicky calculated the velocity dispersions explicitly, arriving at 80 km/s as the expected value. The observed velocity dispersion, however, was 1000 km/s. From this, he could also calculate the mass-to-light ratio and arrived at the same conclusion as Lundmark:



Figure 1: Image of the Coma Cluster, an important player in the discovery of the DM problem. Image taken by an amateur [4]. Public domain.

there has to be much more DM present in the universe than expected [1].

The next great step came in the form of galactic rotation curves, a schematic example of which can be seen in Fig. 2. Galactic rotation curves show the orbital speed of stars in a galaxy as a function of distance from the galactic center and they can be used to probe the mass distribution of a galaxy. While not unknown beforehand, their use took off after the advent of radio astronomy in the 1950s and 1960s. Initial studies mostly focused on the galactic centers. Rubin and Ford were the first to measure a rotation curve all the way to the edge of a galaxy, with the use of spectroscopy [1][6][7]. They observed a flat rotation curve, as seen in Fig. 3, meaning that the outer stars moved with the same velocity as the inner ones. Others, such as Whitehurst and Roberts, began to make similar observations using radio measurements [8]. In 1978, Rubin, Ford, and Thonnard published a further 10 measured rotation curves of spiral galaxies - all flat [9]. By the end of the decade, the evidence in favor of DM was overwhelming, and thus attention shifted towards figuring out what it is exactly [10].

2.1 Possible solutions

Once the evidence in favor of dark matter became impossible to ignore, physicists started to look for solutions. Initially, the most obvious was that it was simply made up of faint stars, planets, or other such compact objects that were too faint to observe [1]. Gravitational microlensing enables the detection of faint objects. Predicted by general relativity, the effect is caused by light having to travel through a section of space-time that is bent by a massive object. An observer on Earth attempting to take an image of the source of the light (e.g. a star) would receive a distorted image due to the effect. Many faint or dark objects were found using this effect, but their mass was not nearly enough account for the missing mass in

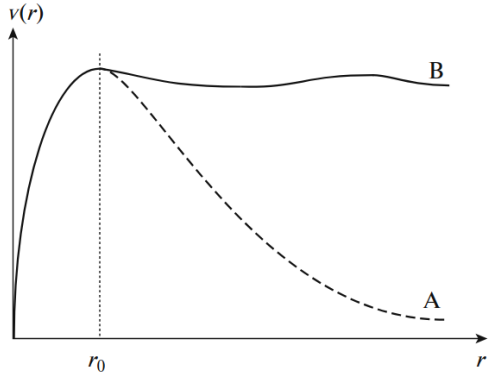


Figure 2: Illustration of a typical galaxy rotation curve. r is the distance from the galactic center, while $v(r)$ is the rotation velocity. (A) shows the expected curve based on only the visible stars, while (B) shows the observed curve. r_0 shows the border between the inner and outer regions of the galaxy. Source: [5]

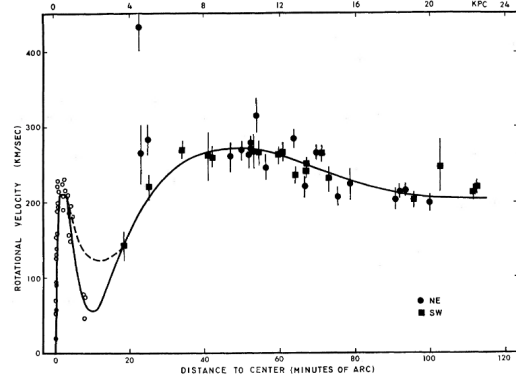


Figure 3: Rotation curve of the M31 galaxy, as measured by Vera Rubin and Kent Ford. Defying expectations, the velocity of stars in the outer region does not drop off, and the curve remains stubbornly flat. This is one of the earliest explicit observational proofs in favor of DM halos. Source: [6]

our galaxy.

Nowadays, the most widely-researched theories are all about particle DM. A DM particle (or particles) would have to satisfy three main criteria:

- have mass,
- be stable,
- and have no interaction with photons.

Initially, SM neutrinos seemed to be promising candidates as they checked all three boxes. However, due to their light weight, they would have had highly relativistic energies in the early Universe, when the large-scale DM structures were formed. Simulations revealed that due to this high energy, neutrinos could not have formed the observed DM halos, signaling the end of their candidacy [1].

Even though no other Standard Model particles fulfill the three DM conditions, physicists continued to come up with new models. Supersymmetry (SUSY), first emerging in the 1970s, is one such theory. SUSY gets its name from the idea that a symmetry exists in nature between fermions and bosons. This means that every SM particle has a “twin” with the same quantum numbers but opposite spin characteristics. In other words, every half-integer spin fermion must have an

integer-spin boson pair, and vice versa. Many SUSY particles have been evaluated as DM candidates [1].

2.2 Thermal relic dark matter

Now inundated with possible DM candidates, it would make sense to organize them based on their theorized properties. One usual method of classification is based on their masses. The only “true” constraints on DM mass come from astrophysics. A particle with mass below the lower limit, 10^{-22} eV, would have a Compton wavelength that conflicts with Cosmic Microwave Background observations. Meanwhile, the upper limit for DM is around tens of solar masses, coming from the observed stability of other (visible) objects in dark matter halos. Simply put, DM candidates above the limit would have such high masses, that they would destabilize solar systems with their gravitational effects. These limits leave us with a staggering 90 orders of magnitude of possible DM masses [11][12].

This range is based on what we know for certain about dark matter. However, if we make a few simple assumptions, we can make our jobs considerably easier. A common assumption, called “thermal relic dark matter”, relates to the origin of DM. The assumption is, that at some point in the early universe, DM and SM particles were in thermal equilibrium and could freely turn into each other. This assumption would indeed be true as long as there is some non-gravitational coupling between SM and DM, and it is larger than the expansion rate of the universe [13]. However, as the universe cooled, the interaction rate lowered and the abundance of DM was “frozen”, and has remained unchanged to this day.

Based on these assumptions, a minimum annihilation cross section for dark matter can be calculated:

$$\langle\sigma v\rangle = 10^{-26}\text{cm}^3/\text{s} \quad [1] \quad (1)$$

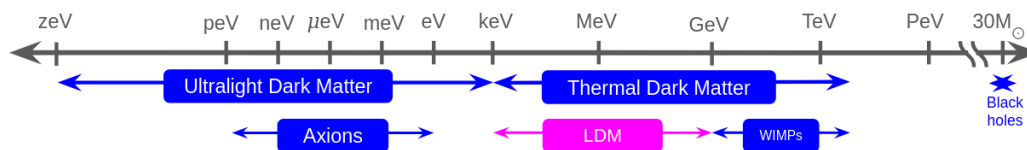


Figure 4: Graphic showing some DM candidates, organized based on their mass. Light Dark Matter, the region of interest for LDMX is highlighted in magenta. Figure inspired by Fig. 1. in Ref. [11].

This is fortunate for two reasons. One, DM with this annihilation cross section can only have mass between a few keV and around 100 TeV, giving a substantially smaller range to search. Two, it is not zero - therefore, production of dark matter at accelerators is possible.

It could also be fortunate for a third reason: it is deceptively close to the number that would arise from the weak force. Thus, a new theory was catapulted to fame: “weakly interacting massive particles”, or WIMPs. It is an attractive theory, since no new force would need to be introduced, and it fits neatly into the upper half (GeV and above) of the aforementioned keV-TeV range. This region has been subject to multiple searches since this “WIMP miracle” was articulated, but they have remained fruitless [14].

However, the lower range of thermal DM, often called “light dark matter (LDM)”, still remains largely unexplored [11]. LDMX aims to explore exactly this sector.

2.3 A benchmark model for designing LDMX: Dark bremsstrahlung

The benchmark model for LDMX is a hidden sector DM model, where DM couples to the SM through a new low-mass mediator particle, the dark photon (A'). Fig. 5 shows the production method for DM. This model introduces four free parameters: the mass of the dark photon ($m_{A'}$), the mass of dark matter (m_χ), the coupling constant between A' and Standard Model particles (ϵ), and the coupling constant between A' and the dark matter particles (α_D).

LDMX’s benchmark assumes that the mass of the final-state DM particle is below 1 GeV, and that the mass of the mediator is at least twice of it ($m_{A'} > 2m_\chi$). In this case, the decay of the dark photon into a DM pair is not kinematically forbidden, making that the main channel for dark photon decay.

LDMX will attempt to produce dark matter through a process called “dark bremsstrahlung”, where electrons are scattered off a heavy nucleus to produce dark photons, which then decay to dark matter. Fig. 7 illustrates the dark bremsstrahlung process. Thus, the dark matter signal would be invisible in our detectors.

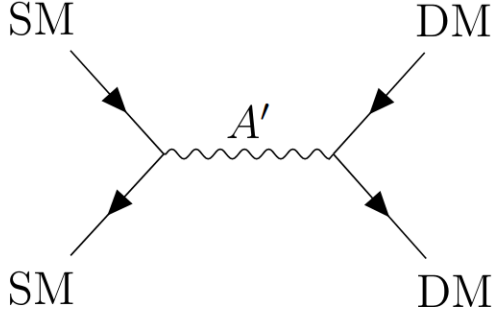


Figure 5: Feynman diagram showing the Standard Model (SM) particles annihilating to dark matter (DM) through the mediator dark photon (A').

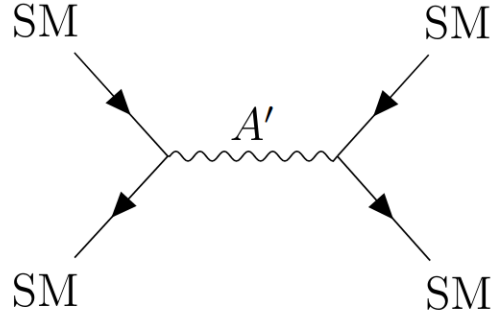


Figure 6: Feynman diagram showing the possible visible signal of dark matter production, in which Standard Model (SM) particles annihilate through a mediator dark photon (A') to further SM particles.

2.3.1 Visible signature

The dark photon also has a chance of decaying back into SM particles. The exact likelihood of this happening depends on the relationship between the mass of the dark photon and the mass of the DM particles. The different possible mass relations are the following:

- $m_{A'} > 2m_\chi$: the visible decay channel is suppressed due to the small coupling between the SM and the dark photon. Nevertheless, it still has a chance of happening.
- $m_{DM} < m_{A'} < 2m_\chi$: the decay of dark photons into DM is kinematically forbidden, making the visible decay channels more viable.
- $m_{A'} < m_\chi$: The mediator only decays visibly. This decay channel is also known as secluded annihilation.

The visible signature in all cases would be an electromagnetic (EM) shower in the ECal from the recoiling electron and another displaced EM shower in the ECal or HCal, with no track leading to it. Lighter dark photons have longer expected lifetimes and have a higher chance of the decay occurring in the main HCal [13].

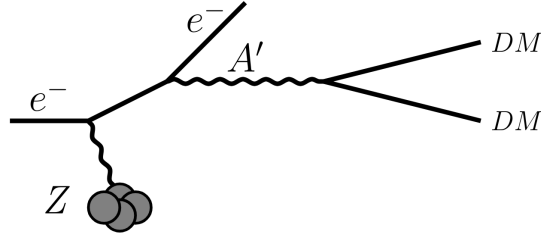


Figure 7: Schematic view of dark bremsstrahlung. A standard model electron (e^-) scatters off a nucleus, producing a dark photon (A'), which decays into dark matter (DM).

3 The Light Dark Matter eXperiment

LDMX will be a fixed-target missing momentum and missing energy experiment. It will be the primary user of the new Linac to End Station A (LESA) beamline, currently under construction at SLAC. LESA will be a parasitic beamline of the LCLS-II linear accelerator, and will extract $\sim 60\%$ of unused electron bunches from it to deliver them to End Station A (ESA), where the LDMX detector will live [15]. The layout of LESA and its surroundings can be seen in Fig. 8. To LDMX specifically, LESA will deliver 8 GeV electrons at a roughly 37 MHz frequency, which means that LDMX will be able to reach $\sim 10^{14}$ electrons on target within 1-2 years with one electron per bunch [16].

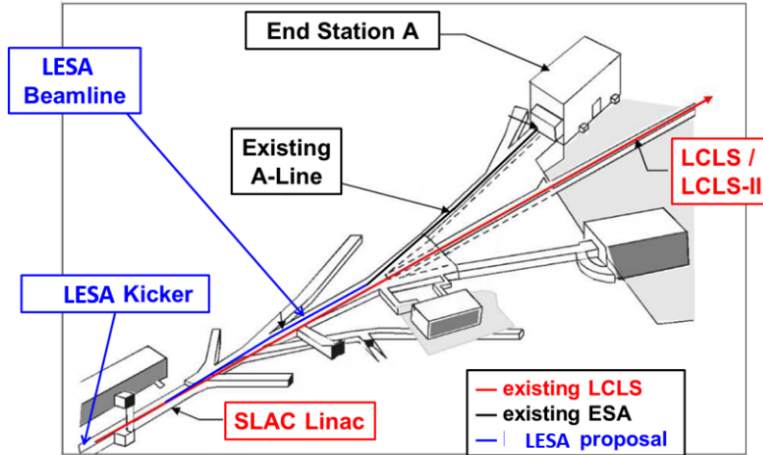


Figure 8: Schematic layout of the LCLS-II and End Station A. Source: [15]

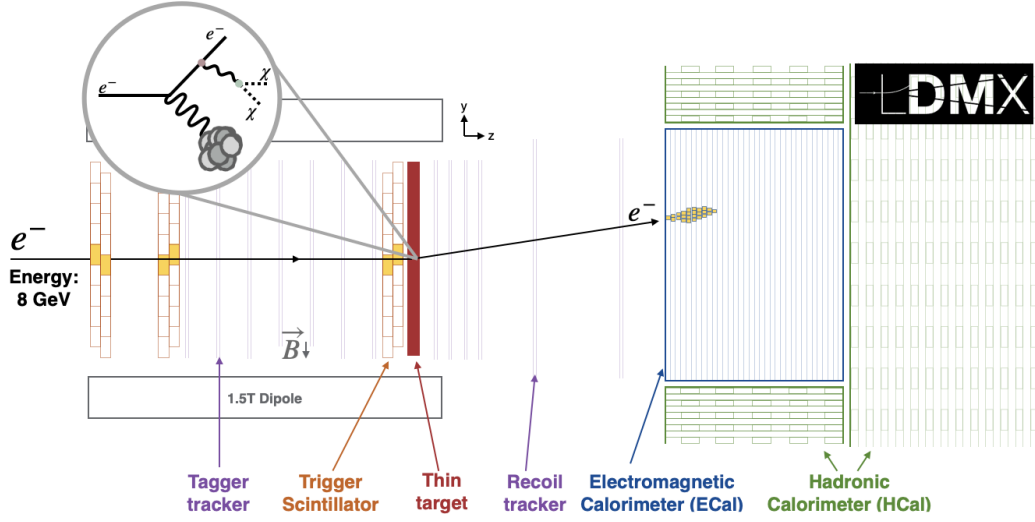


Figure 9: Schematic view of the full LDMX detector setup and DM signal. Source: [16]

A schematic view of the full detector setup can be seen in Fig. 9. The electrons will scatter off a thin tungsten target and hopefully produce DM. The dark bremsstrahlung benchmark model signal is a single electron, with missing energy, a large transverse momentum kick, and nothing else, since both dark photons and DM are invisible to detectors. To record this signal, the LDMX detector includes trackers both before and after the target, as well as an electromagnetic calorimeter. A trigger scintillator is also included for triggering. Additionally, as LDMX is designed to handle one electron on target at a time, the trigger scintillator will be used to control the number of electrons in the case of pileup. Finally, a hadronic calorimeter is placed around and behind the ECal.

The main background process for dark bremsstrahlung is SM bremsstrahlung, resulting in a photon alongside the recoiled electron or the electron passing through the target without interaction. Both of these signals can be effectively vetoed with just an ECal. However, the photon from the bremsstrahlung could further decay into hadrons, or other electro-nuclear and photo-nuclear processes could result in the production of neutral particles that would escape detection in the ECal. The HCal has a critical role in enabling us to veto these events. This work is mainly concerned with the HCal, which will now be described in detail.

A simulated view of the full HCal can be seen in Fig. 10. It will have two main parts, the side HCal (around the ECal) and the main HCal (behind the ECal). The HCal is a sampling calorimeter, made from alternating layers of 25-mm thick steel absorber plates and plastic (polystyrene) scintillator bars. The scintillator bars

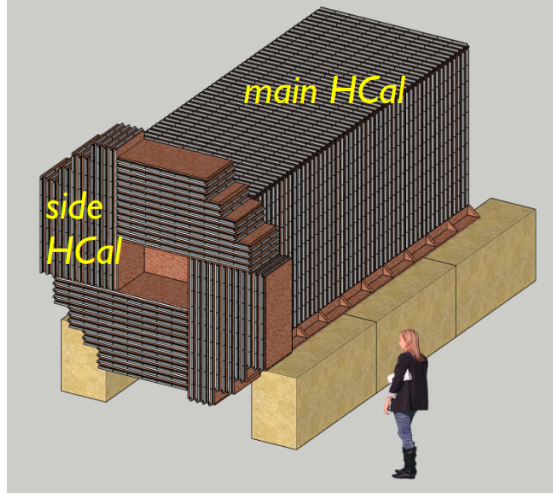


Figure 10: Simulated view of the full HCal, with the side HCal and the main HCal noted. Source: [18]

are 2-m long, 20-mm thick and 50-mm wide. They have alternating orientations layer-by-layer. If a high-energy particle passes through, the bar emits photons through scintillation. This is called a “hit”. These photons are gathered up by a wavelength-shifting fiber, which then carries them to silicon photo-multipliers (SiPMs) at the two ends of the bar.¹ Four scintillator bars are glued together to form one quadbar. One Counter Mother Board (CMB) per quadbar holds the SiPMs and provides connection for the HDMI cable that transmits the signal to the High Granularity Calorimeter Read Out Chip (HGCROC) board. One HGCROC board will read out signals from 16 quadbars (64 channels).

The final HCal will consist of 96 layers of 40 scintillator bars each, resulting in a final size of $2 \times 2 \times 4$ meters [17]. A smaller-scale prototype was made to test this design.

3.1 The hadronic calorimeter prototype

The HCal prototype is a scaled-down version of the main HCal. The main difference between the prototype and the final version is the size - the prototype is only 19 layers deep, with two quadbars in the front nine layers and three quadbars in the back ten layers. A schematic drawing of the prototype can be seen in Fig. 11, while a picture taken during test beam can be seen in Fig. 12. The prototype used the same readout chain as the final experiment, although the final

¹This means that there are two readout channels associated with each bar in the backHCal. In the sideHCal, due to space constraints, the readout is single ended [14].

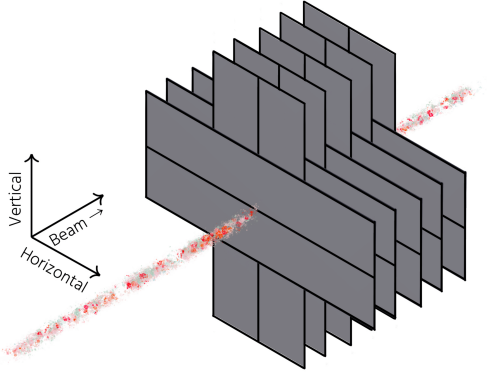


Figure 11: Schematic drawing of the HCal prototype [19]



Figure 12: The HCal prototype at the 2022 CERN testbeam [19]

LDMX HCal will use an improved version of the HGCROC board based on lessons from this test beam with the v2 HGCROC board. The specific workings of the HGCROC chip are important to contextualize some of the results presented later in this thesis.

The HGCROC chip was originally designed to readout the High Granularity Calorimeter (HGCAL) of the Compact Muon Solenoid (CMS) experiment at CERN [20]. In preparation for the High-Luminosity LHC, the HGCAL will replace the endcap calorimeters currently installed in CMS. The chip can handle a total of 72 readout channels, although the LDMX HCal only uses 64. A simplified schematic of the different components in each readout channel can be seen in Fig. 13 and these highlights are reviewed here. For a full description, please visit reference [20].

The signal readout from the SiPM goes first into the preamplifier (PA). The PA integrates the incoming charge into a pulse with peak signal proportional to the incoming charge. The proportionality is governed by the PA gain, which is a configurable variable [20][21].

Then, the amplified signal is sent to three different components. It is always sent to the analog-digital converter (ADC) through a pulse shaper (SH). The ADC converts the analog signal to a digital one by sampling it at 25-ns intervals, taking a total of eight measurements. This sampling frequency (≈ 40 MHz) was chosen originally to match the frequency of bunch crossings in the LHC.

The ADC is capable of holding 10 bits of information, which means that at large enough pulses, it can saturate. The PA can also enter a non-linear regime at large pulses, making the use of ADC measurements non-viable, even if the ADC did

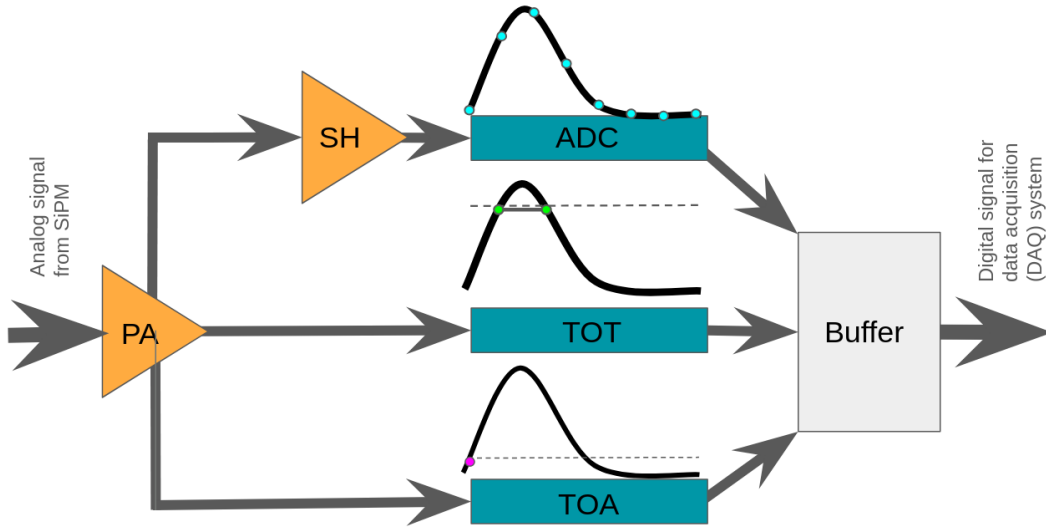


Figure 13: A simplified schematic view of the “inside” of a readout channel in the HGCROC chip and its signal processing paths. The ADC takes eight samples of the pulse at 25-ns intervals. The TOT measures the time the pulse spends above the TOT threshold (marked with a dashed line). The TOA measures when the pulse hit TOA threshold (marked once again with a dashed line) on the rising edge. Figure inspired by Fig. 1. in Ref. [21].

not saturate. For this purpose, a time measurement is used. The time the pulse spends over this saturation line is measured by a time-to-digital converter (TDC) to provide the time-over-threshold (TOT) value. This stays linear at large pulses, extending the range of measurements. TOT is not measured if the pulse does not reach a minimum threshold, which is set to be slightly below the saturation line [20][21].

The third possible measurement is time-of-arrival (TOA). This measurement is taken if the signal reaches a minimum threshold to be considered a pulse. Then, a TDC measures the rising edge of the pulse with high precision, to provide pulse timing information.

3.2 2022 CERN test beam

The test beam was conducted at the T9 beam line in the East Experimental Area, pictured in Fig. 14. The T9 beam line is a mixed secondary beam, which means that before the beam reaches the experimental area, it interacts with a target. This results in a beam that contains a mixture of hadrons and electrons. Muons may also be created by inserting collimators that block other particles. Identifying and

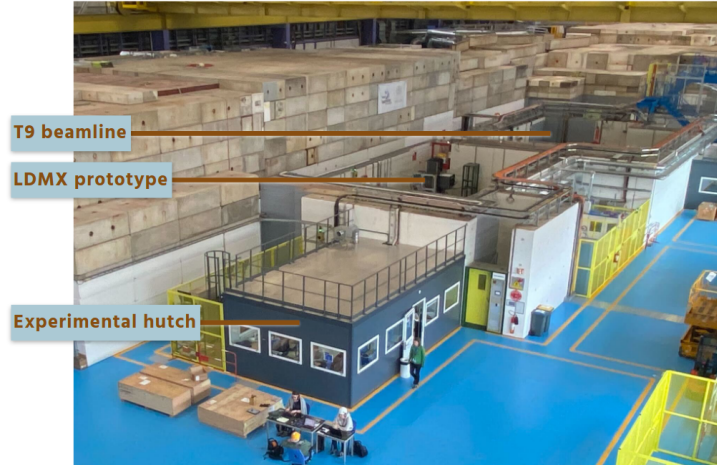


Figure 14: A picture from test beam of the T9 beam line and the LDMX HCal prototype. Source: [23]

discriminating between particles is made possible through two Cherenkov detectors installed on the beamline [22].

While the T9 beam line is capable of delivering particles with energies up to 15 GeV, data at test beam was only taken with particle energies up to 4 GeV [24]. Specifically, data was taken with:

- 4 GeV defocused muons
- 0.3-4 GeV electrons
- 0.1-4 GeV pions.

The full list of runs is available on GitHub.² This list excludes calibration runs, such as the ones necessary for the ADC-TOT cross-calibration mentioned in Sec. 4.2. Data was also taken with no beam present in order to establish the baseline, also referred to as the “pedestal”, for the HCal electronics. It is calculated by fitting a Gaussian to the distribution of recorded ADC measurements. The mean of the Gaussian is the pedestal, while its standard deviation (σ) characterizes the pedestal’s noise level [25][17]. The pedestal plays an important role in event selections.

²<https://github.com/LDMX-Software/hcal-testbeam-analysis/blob/main/runs.md>

4 Methods and Analysis

To detect visible DM signatures as described in Sec. 2.3.1, the ability to distinguish between hadronic and electromagnetic showers is critical. To this end, electromagnetic showers are studied and characterized in this thesis. Specifically, electron response linearity, electron energy resolution, and lateral and longitudinal shower shapes were extracted from the data taken at the 2022 CERN test beam with the HCal prototype.

4.1 Event selection

Electron runs with energies ranging from 300 MeV to 4 GeV were used. Only bars registering hits 5σ above the pedestal values were included in the calculations. Due to suspected pion contamination, events with hits in the back-most five layers were removed. Because the HCal prototype includes a steel absorber plate before the first scintillator bar layer, some electrons showered before reaching the scintillator bars. This leads to a problem: if the particle showers before it reaches the first scintillator layer, we do not see the incident particle. This can lead to inconsistent energy reconstruction results, as we cannot distinguish between particles that showered in the first steel layer from those that showered upstream of the prototype. Therefore, events with multiple hits registered in the first layer were discarded. This event selection was used for both the energy resolution calculations and the shower shape analysis.

4.2 TOT energy deposition

The main measurements in this analysis is the ADC and TOT measurements. As mentioned in Sec. 3.1, the TOT takes over for the ADC past saturation. To reconstruct the energy deposited during the time the signal spends over the saturation level, cross-calibration has to be performed between ADC and TOT measurements. Crucially, at the time this work is being written, this calibration has not been done yet. Practically, this means that if a charge is deposited into a scintillator bar which saturates its associated ADC, some deposited energy is not recovered.

4.3 Data structure

The data taken at test beam were recorded originally in *ROOT* [26] and were decoded into *csv* files using code available on GitHub as part of LDMX Software [27]. The files could then be analyzed using *Python*, specifically with the packages *numpy* [28], *matplotlib* [29], *scipy* [30], and *pandas* [31][32]. The code I wrote to

perform the analyses presented in this thesis can be found on GitHub³. One line of data, representing a single event as an example, can be seen in Fig. 15.

pf_event	72362.0
tot_end0	680.0
toa_end0	31.2
adc_sum_end0	3722.0
adc_mean_end0	465.2
adc_max_end0	1023.0
layer	1.0
strip	4.0
pf_spill	122.0
pf_ticks	1613779.0
tot_end1	1000.0
toa_end1	34.5
adc_sum_end1	3667.0
adc_mean_end1	458.4
adc_max_end1	1023.0

Figure 15: One line of data recorded in a 4-GeV electron run. This would normally be one row in the database; it was only transposed for demonstrative purposes.

The different fields are:

- **pf_event**: The event number. One event is one trigger, hopefully corresponding to one electron shot into the prototype.
- **tot_end0(1)**: TOT measurement for one end of a bar. It is 0 if the TOT threshold was not reached.
- **toa_end0(1)**: TOA measurement for one end of a bar. It is 0 if the TOA threshold was not reached.
- **adc_sum_end0(1)**: sum of all eight ADC measurements in one end of a bar.
- **adc_mean_end0(1)**: mean of all eight ADC measurements in one end of a bar.
- **adc_max_end0(1)**: the highest of the eight measured ADC samples in one end of a bar.

³<https://github.com/gajdangreta/hcalTestBeam>

- **layer**: The index of the layer in which the event was recorded. One-indexed.
- **strip**: The index of the scintillator bar in which the event was recorded. Zero-indexed.
- **pf_spill** and **pf_ticks**: these provide additional timing information necessary to synchronize the front and back halves of the prototype. They are not necessary for the calculations going forward.

Let us unpack the values from the example in Fig. 15. The two ends of a scintillator bar are treated completely separately on the level of the electronics, as described in Sec. 3.1, meaning each end serves as one channel. That means that their ADC values, pedestal values, noisiness, etc. can be entirely different. We would expect that after the subtraction of pedestals, the two values would become largely the same. The variation in TOA values is due to the hit position's offset along the bar from the exact middle, so the signal has to travel just a bit longer to get to one side than the other. The difference in the two TOT values is interesting, as it is not expected. It could be an error in TOT decoding. However, as was mentioned in Sec. 4.2, an ADC-TOT calibration is not available at the time of writing this thesis. As a result, TOT values do not play any role in the analysis.

The pedestal values, as well as the values needed to convert the ADC units to energy, are generated from calibration runs and are stored in separate files.

4.4 Physics of shower development

Particles can have multiple distinct reactions while traveling through matter, depending on particle species and energy. Through these interactions, an incident particle will lose energy until it exits the material or is fully absorbed. In the case of high-energy particles, a shower of secondary particles may result as a consequence of the interaction. If this shower takes place in a calorimeter, as illustrated in Fig. 16, we can capture a part of its energy and describe its shape. The shower shape depends on the species and energy of the primary particle, which means that we can identify an unknown particle in a calorimeter based on its shower profile.

For high-energy (>100 MeV) electrons and positrons, the dominant process for energy-loss is bremsstrahlung [34]. Through interacting with the electromagnetic field of a nucleus, photons are radiated off, which then can pair-produce further electrons and positrons. This chain reaction then continues until the energy of the primary particle runs out, and no further particle production can occur. This “critical energy” is different for photons and electrons. Fig. 17 shows the different processes through which particles can lose energy traveling through matter and

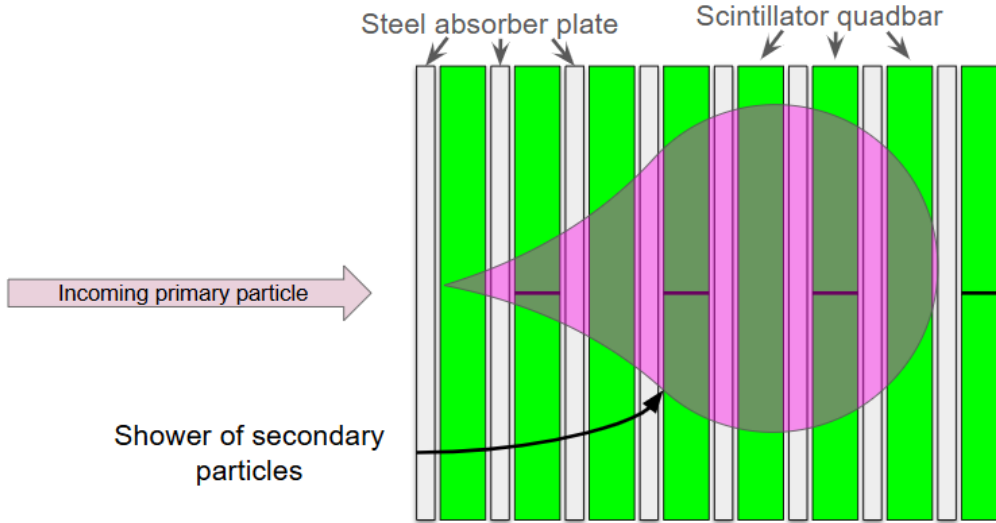


Figure 16: Schematic view of a particle shower. The front half of the HCal prototype is shown, with the scintillator quadbars marked in lime green. Based on a figure in Slide 8 from Ref. [33].

highlights the location of the critical energy for both electrons and photons. It can be seen that, while they are not the same, they are close enough that they do not need to be distinguished during later calculations.

For photons, two factors contribute to the critical energy. On the one hand, pair-production simply cannot happen if the energy of the photon is below the combined mass of an electron and a positron (2×511 keV). However, Compton scattering usually takes over as the dominant process at higher energies [34].

For electrons, the “stopping point” for further particle creation is the critical energy, E_C . It is dependent on the material through which the electron is passing. Above E_C , bremsstrahlung is dominant, while below ionization wins [34].

After this point, the rest of the energy is slowly absorbed, and no more particles are created. Therefore, an EM shower increases in energy deposition until a shower maximum is reached, after which it decreases as the remaining energy is fully absorbed [34].

4.5 Energy response and resolution

If the dark sector mediator does indeed decay in the HCal, we would like to know its energy, in addition to its observed decay products. By knowing the species and

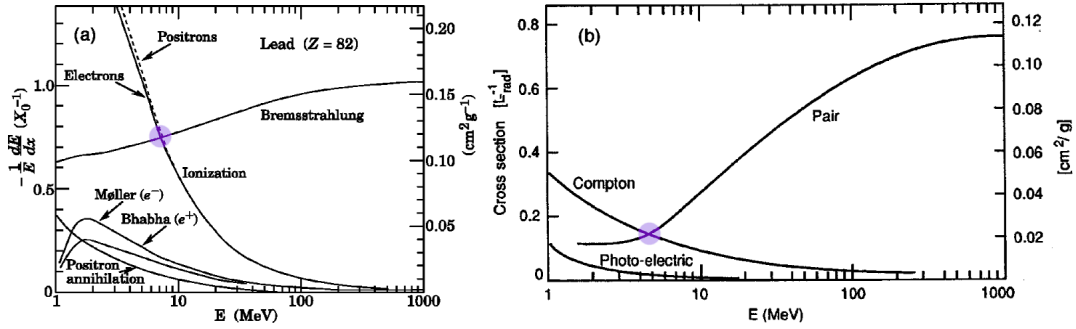


Figure 17: On the left (a), the fractional energy loss of electrons (and positrons) in lead is shown as a function of their energy. On the right (b), the cross-section of different photon interactions in lead are shown as a function of energy. The critical energy is highlighted in purple on both plots. Source: [35]

the energy of the decay products, their momentum may be calculated. Due to the laws of energy and momentum conservation, the energy and momentum of the dark photon is then known, meaning we know the mass of the mediator. Of course we cannot measure any of these values *exactly*, only with some finite precision. Nevertheless, it would allow us to at least place some constraints on dark photon mass. If visible and invisible signatures were both present, based on their ratio (as described in Sec. 2.3.1), we could also say something about the ratio of $m_{A'}$ and m_χ , and therefore about the value of m_χ . Establishing the HCal's energy response and resolution to different particles are therefore necessary.

The energy response of a calorimeter describes the relationship between the signal the calorimeter produces and the energy of the particle that caused the signal [34]. In other words, it is the average “height” of the signal, divided by the incoming energy. Characterizing this response is different for homogeneous and sampling calorimeters.

Homogeneous calorimeters are made entirely of active material. The particles are fully absorbed in this active material, all of their kinetic energy contributing to signal generation. These calorimeters typically have a high energy resolution, and a large fraction of the deposited energy can be recovered [34].

Our HCal is not a homogeneous calorimeter, but a sampling one. The layers of active material (the scintillator bars) are interposed with a passive absorber (steel), as described in Sec. 3.1. The absorber is what absorbs most of the particles' energy and induces the particle showers, while the active material only “samples” the released energy. While this leads to lower energy resolution due to sampling fluctuations, the segmentation allows for the measurement of particle location (and

therefore shower shapes, see Sec. 4.6), as well as for a more compact size and lower costs [34].

The energy response of a sampling calorimeter, among other things like sampling frequency, is governed by its “sampling fraction”. The sampling fraction is the energy deposited into the active layers by a minimum ionizing particle (MIP) relative to the total energy deposited into the detector [34].

A MIP would lose around 11.4 MeV/cm in iron⁴ and around 2 MeV/cm in polystyrene. This means, that in the HCal prototype, a MIP would lose 28.5 MeV in the 2.5-cm-thick steel absorber and 4 MeV in the 2-cm-thick scintillator bar. The sampling fraction is therefore:

$$\frac{4 \text{ MeV}}{28.5 \text{ MeV} + 4 \text{ MeV}} = 0.123 \quad \rightarrow \quad 12.3\% \quad (2)$$

This is, however, not the actual percentage of energy deposited by particles. On the one hand, MIPs are theoretical particles. They are usually approximated by 4-GeV muons, but even they deposit more energy than MIPs due to relativistic effects. In the case of electromagnetic showers, the response to electrons will be lowered compared to MIPs. If the average atomic numbers of the absorber and the active material are different, low-energy photons will have a higher cross section for interaction in the passive material and therefore have a lower chance of contributing to the signal in the active layers [34]. Still, this calculation gives us a ballpark in terms of how big of a signal we can expect in the HCal as a response to a particle hit.

We would naturally like this response to scale with the energy of the incoming particle. This is called response linearity. In theory, the response to electrons should always be linear. Since electrons use up all their energy while showering (as discussed in Sec. 4.4), a 4-GeV electron should induce a signal four times bigger than a 1-GeV electron. In practice, non-linearity can happen as a result of instrumental effects or miscalibration [34].

While the energy response is the signal “height” divided by incoming energy, the energy resolution is the “width”. It gives a measure of how certain an energy measurement of a particle is. The main limiting factor for resolution in sampling calorimeters is its sampling nature. Because not all of the energy is collected by the active material, the amount of energy collected is vulnerable to statistical fluctuations. Instrumental effects, such as electronics noise, also contribute [34].

⁴Steel is mostly iron, and its characteristics are close enough that the numbers for iron may be used in the place of steel.

The resolution of a calorimeter (for EM showers) is generally described by the equation:

$$\frac{\sigma(E)}{E} = \frac{a}{\sqrt{E}} \oplus \frac{b}{E} \oplus c, \text{ i.e. } \sqrt{\left(\frac{a}{\sqrt{E}}\right)^2 + \left(\frac{b}{E}\right)^2 + c^2} \quad (3)$$

where $\sigma(E)$ is the signal width, E is the energy of the incoming particle, and a, b and c are free parameters. $\oplus$ denotes a quadratic sum. The first term is the stochastic term, which includes the sampling fluctuations previously described. The second term is the noise term, coming from electronics effects. The third constant term contains all other effects that do not depend on particle energy, which are instrumental effects like the impacts of detector geometry or radiation damage [35].

The electron energy response and resolution were determined for the HCal prototype using test beam data. After the event selection, described in Sec. 4.1, the deposited energy was summed up and converted into MeV from ADC units for each event. As described in Sec. 4.3, we expect the two ends to have essentially the same results after pedestal subtraction is performed. By treating the two ends separately, we can see if there are any differences that need to be addressed.

The deposited energy sums were used to produce distributions, which were then fitted with Gaussian functions. By plotting the means of the Gaussian fits against the beam energy, and fitting the resulting data point with a line, we obtain the energy response function. Conversely, by dividing the standard deviation of the fits with the mean, and fitting the results with the function described in Eq. 3, we obtain the energy resolution.

4.6 Shower shapes

The location of this shower maximum (described in Sec. 4.4) can be calculated. Let us simplify the model to only include the two dominant processes: pair production for photons and bremsstrahlung for electrons [36].

An electron with the energy E_0 will, on average, undergo bremsstrahlung to produce a photon and electron with $E_0/2$ energy each after one radiation length, X_0 . X_0 is the distance after which a particle's energy is reduced by a factor of $1/e$. The radiation length depends on the material. Meanwhile, a photon will cover a mean path of $9/7 X_0 \approx X_0$ before pair producing [34][36].

This means, that effectively after $2X_0$, there are four particles with $E_0/4$ energy each in the detector, eight with $E_0/8$ after $3X_0$ and so on. A drawing of such a simplified cascade can be seen in Fig. 18. In reality, this is often not so simple. Often, multiple photons are produced by bremsstrahlung, and they usually only

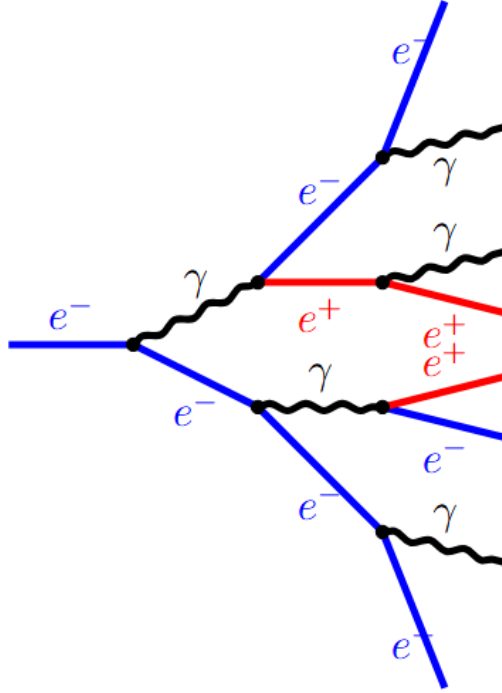


Figure 18: A schematic drawing of a simplified electromagnetic cascade induced by a high-energy electron. Electrons, positrons, and photons are marked with red, blue, and black, respectively.

carry away a small portion of the energy each (few MeV) [34]. This calculation produces a close enough approximation to reality for interpreting the shower shapes made from test beam data [36].

If an electron's or photon's energy falls below the critical energy E_C , it is absorbed by the material rather than producing more particles. Therefore, the location of the shower maximum will be proportional to E_0/E_C . This means that after distance d (in units of X_0):

$$\text{Number of particles: } n(d) = 2^d \quad (4)$$

$$\text{Energy of particles: } E(d) \approx \frac{E_0}{2^d} \quad (5)$$

At the shower maximum d_{max} :

$$\text{Number of particles: } n(d_{max}) \approx \frac{E_0}{E_C} \quad (6)$$

$$\text{Shower maximum: } d_{max} \approx \ln \frac{E_0}{E_C} \quad (7)$$

Let us do the calculations for the test beam data taken with the HCal prototype. As described in Sec. 3.1, the prototype is made of alternating layers of steel absorber plates and plastic scintillator bars. The bars are made of polystyrene, which has a radiation length of 41.31 cm. This means that one $d = 2$ cm thick bar has a radiation length of:

$$X_0^{bar} = \frac{d}{X_0} = \frac{2 \text{ cm}}{41.31 \text{ cm}} = 0.05 X_0. \quad (8)$$

Compare this to the steel absorber:⁵ The radiation length of steel is 1.76 cm. Since the absorber plates of the HCal prototype are $d = 2.5$ cm thick, one layer has the following radiation length:

$$X_0^{steel \text{ layer}} = \frac{d}{X_0} = \frac{2.5 \text{ cm}}{1.76 \text{ cm}} = 1.42 X_0. \quad (9)$$

Based on this, the scintillator bar's contributions to the radiation length of one layer are negligible, therefore only the steel was used in the calculations.

Steel's critical energy for bremsstrahlung is 22 MeV. Tab. 1 lists the expected shower maxima in our electron runs.

Beam energy	Shower maximum [# of layers]
300 MeV	1.84
500 MeV	2.2
1 GeV	2.69
4 GeV	3.66

Table 1: Predicted shower maxima

One can also study the shower's transverse development by looking at the detectors Molière radius, ρ_M . 90% of a shower's energy is contained in a cylinder with a radius of $1 \rho_M$, and 99% is contained in a cylinder with a radius of $3.5 \rho_M$. Steel has the ρ_M of 1.7 cm, which means that $3.5 \rho_M$ is roughly 6 cm. This very last number is fortunate, since one scintillator bar is 5 cm wide - this means that we would expect almost all of the energy to be deposited into the central two bars.

⁵The properties of steel are roughly equivalent to that of iron. Therefore, in these calculations, the values for iron were used for radiation length, critical energy, and the Molière radius.

The event selection, as described in Sec. 4.1, was performed on the data. The ADC units in each bar were converted into MeV. Then, the energy was summed up by layer (scintillator bar) and divided by the number of events to get the longitudinal (transverse) shower profiles. The data from the two ends were treated separately. As said in Sec. 4.3, we expect the two ends to have consistent results once pedestals are subtracted. By treating them separately, we can see if there are any anomalies between them that must be investigated.

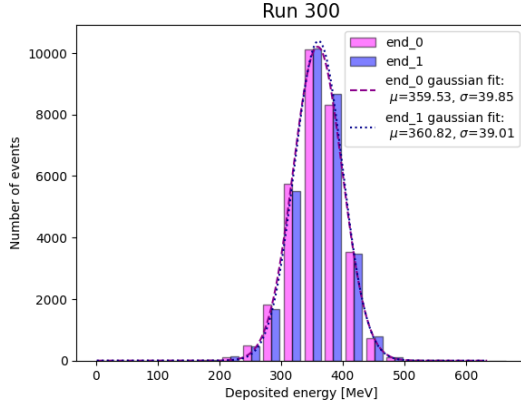
5 Results and discussion

5.1 Energy response and resolution

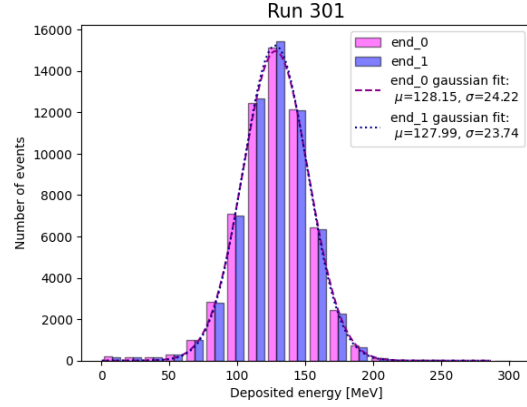
Fig. 19 displays the calculated deposited energies of four electron runs at beam energies ranging from 300 MeV to 4 GeV. As described in Sec. 4.5, the distributions have been fit with Gaussian distributions, treating each end separately. Both the histograms and the Gaussian fits reflect that there is no meaningful or systematic difference between the two ends.

The energy response, seen in Fig. 20, was obtained by following the method outlined in Sec. 4.5. It is especially apparent that the two readout channels of a bar are internally consistent. Due to the lack of ADC-TOT cross-calibration, it is expected that less deposited energy is calculated in the 4-GeV run than what should be. Therefore, the linear fit was done both with and without this run. Excluding the 4-GeV run, on average 13% of the incoming beam energy can be recovered from the HCal prototype. This is surprisingly close to the sampling fraction calculated in Eq. 2, illustrating that this is a reasonable response for the HCal prototype to have.

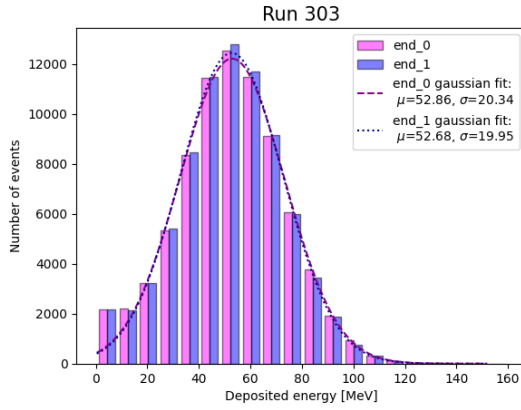
The resolution, seen in Fig. 21, was similarly obtained by following the method outlined in Sec. 4.5. As it became apparent in the earlier steps that the two ends are internally consistent, only one end is shown in the figure. Due to the number of fitting parameters (three) and the number of data points (five), the fit of Eq. 3 could not be performed with the exclusion of the 4-GeV run without destabilizing the fit results. However, the data points (4-GeV run included) are still well described by the fitted function. From the fit parameters, the electron energy resolution of the prototype is $\frac{(8\pm1)\%}{\sqrt{E}} \oplus \frac{(14.2\pm0.1)\%}{E} \oplus (9.8 \pm 0.1)\%$, which is a plausible result. To contextualize the obtained resolution, Tab. 2 contains the sampling fraction and resolution of a few sampling calorimeters in other experiments. The sampling fraction is for MIPs, while the resolution only contains the stochastic term, as that dominates in the energy range important for these experiments [37].



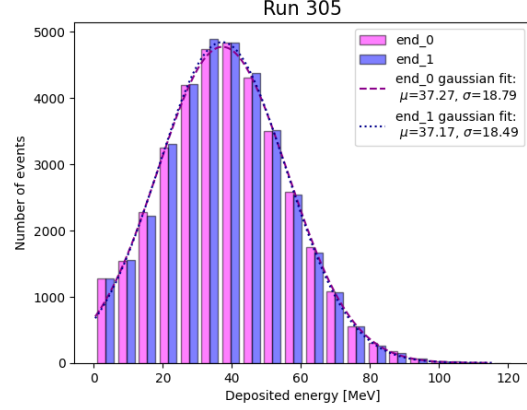
(a) Distribution of deposited energies in a 4 GeV electron run.



(b) Distribution of deposited energies in a 1 GeV electron run.



(c) Distribution of deposited energies in a 400 MeV electron run.



(d) Distribution of deposited energies in a 300 MeV electron run.

Figure 19: Distribution of deposited energies of electron runs. The readouts from the two ends of the scintillator bars are treated separately, seen in pink and purple.

Experiment	Sampling fraction	Resolution
KLOE	17%	$4.7\%/\sqrt{(E)}$
ZEUS	9%	$18\%/\sqrt{(E)}$
NA48	23%	$3.5\%/\sqrt{(E)}$
ATLAS LAr EM barrel calorimeter	$\approx 25\%$	$10\%/\sqrt{(E)}$
PHENIX	29%	$7.8\%/\sqrt{(E)}$
AMS-02	19%	$10.4\%/\sqrt{(E)}$

Table 2: Sampling fraction and resolution of some sampling calorimeters. Only the noise term, which is dominant in the relevant energy range, is noted for resolution. Table adapted from Tab. 2. in [37].

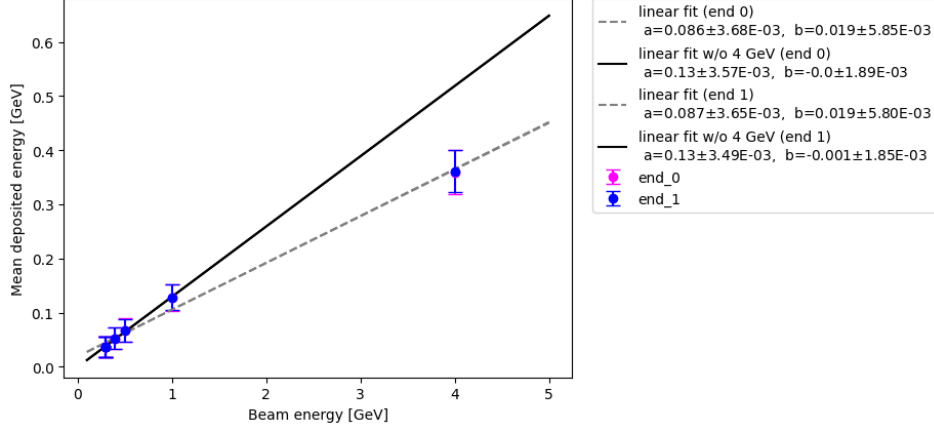


Figure 20: Electron response linearity of the HCal prototype. The solid line fit was made with the 4 GeV run excluded.

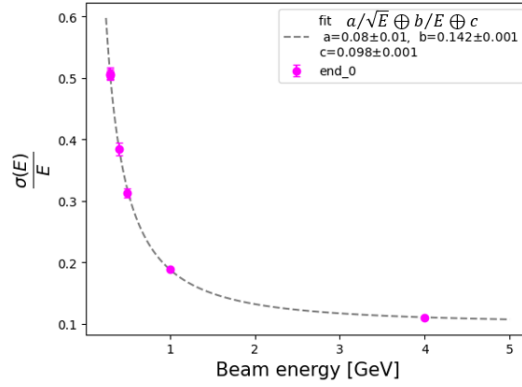


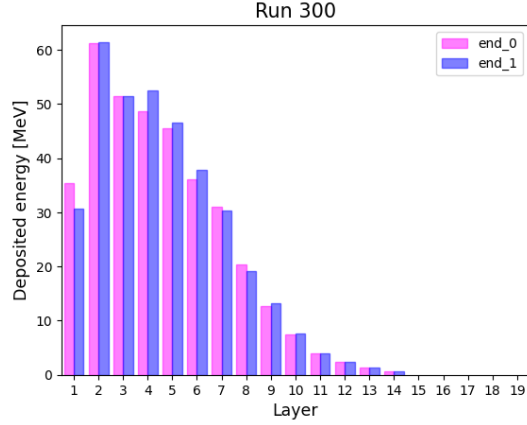
Figure 21: Electron energy resolution of the HCal prototype. Data from only one end of the scintillator bars were used. The uncertainties are all statistical. The irregular error bar of the 300-MeV point is the result of multiple 300-MeV runs with different uncertainties overlapping.

5.2 Shower shapes

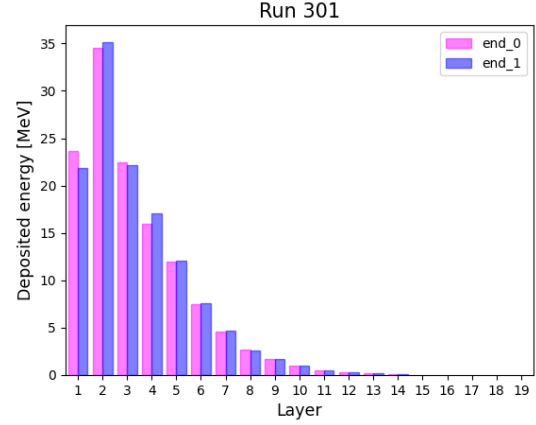
5.2.1 Longitudinal shower shapes

A selected four longitudinal shower profiles can be seen in Fig. 22. They show that the higher the energy of the initial electron, the deeper the shower penetrates the prototype and the “wider” the depth-wise distribution is. This is consistent with the general expectations. It can also be seen that there is no meaningful or systematic difference between the two ends of the scintillator bars (seen in

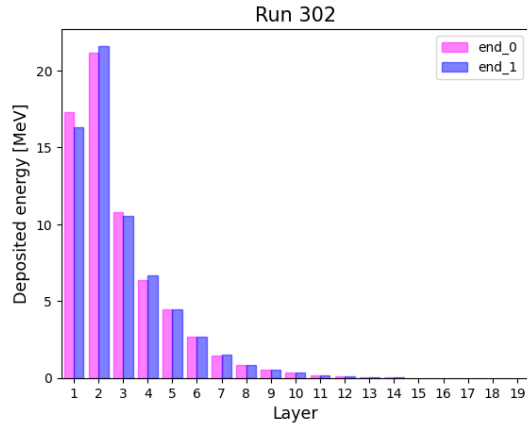
pink and purple). For energies 1 GeV and below, the locations of the observed shower maxima agree with the calculated prediction in Tab. 1. However, a shower maximum in the 4 GeV case is observed in the second layer, while the calculation predicts one in the third. This may be the result of some energy being “lost” (for now) due to the lack of ADC-TOT cross-calibration described in Sec. 4.2.



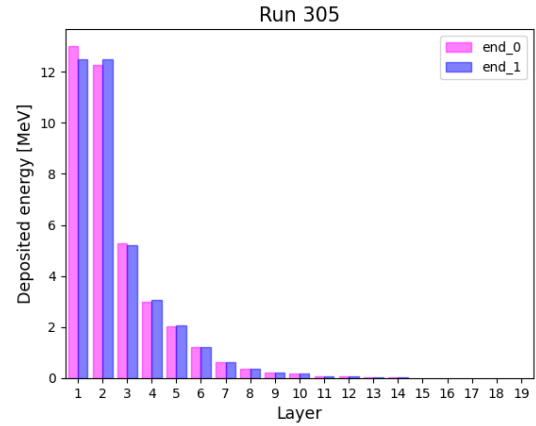
(a) Longitudinal shower profile of an electron run at 4 GeV



(b) Longitudinal shower profile of an electron run at 1 GeV



(c) Longitudinal shower profile of an electron run at 500 MeV

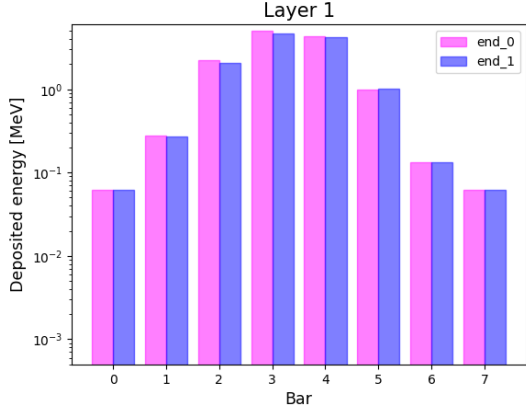


(d) Longitudinal shower profile of an electron run at 300 MeV

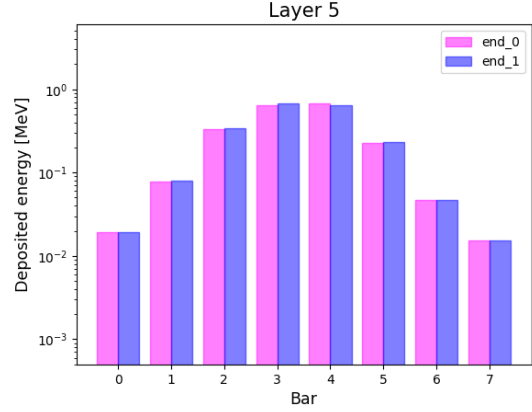
Figure 22: Longitudinal shower profiles of electron runs at different energies. Since readout happens at both ends of the scintillator bars, the two ends are treated separately from each other, seen in pink and purple.

5.2.2 Transverse shower shapes

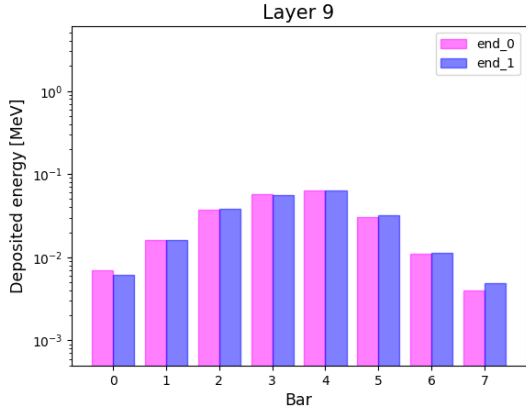
In Fig. 23 four transverse profiles are shown from a 300 MeV electron run. The results agree with the expectation that the EM shower widens in the transverse plane the deeper it goes in the detector. There is once again no meaningful difference between the energies in the two ends of the bars. The result also agrees with the expectations outlined in Sec. 4.6. The most energy is deposited into the two central bars, as they are consistently the most prominent across the layers.



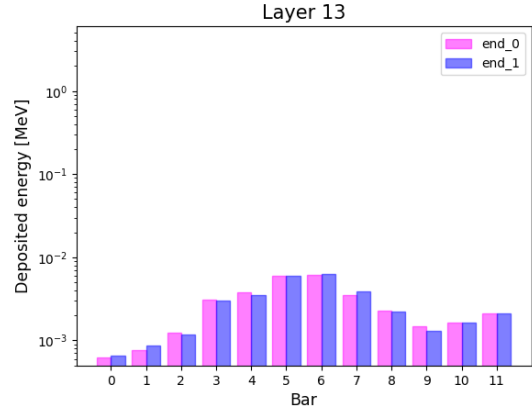
(a) Transverse shower profile of the 1st layer



(b) Transverse shower profile of the 5th layer



(c) Transverse shower profile of the 9th layer



(d) Transverse shower profile of the 13th layer

Figure 23: A few selected transverse shower profiles from a 300 MeV electron run. Since readout happens at both ends of the scintillator bars, the two ends are treated separately from each other, seen in pink and purple. Note the logarithmic scale on the y-axis.

6 Outlook

In conclusion, the results presented here are in line with expectations. The HCal prototype's energy response to electrons is linear up to 1 GeV and its electron energy resolution is in the expected range for a sampling calorimeter. The electron shower shapes observed are also largely consistent with expectations. While these are promising initial results, both for general performance and for visible signature detection, further work needs to be done.

First, the results from the 4-GeV electron run deviate from expectations in both energy response and the longitudinal shower maximum location. The ADC-TOT cross-calibration needs to be done to obtain the correct deposited energy for this run and see whether these deviations persist. If they do, further study of that run will be necessary.

Second, to further establish that the HCal prototype behaves according to expectations, its response to hadrons also needs to be studied. Additionally, the ability to tell hadronic and electromagnetic showers apart is crucial for visible signature detection. Therefore, the characterization of hadronic showers in the HCal prototype is of high importance. This could be done with data taken at the 2022 CERN test beam with pions of varying energy. While there is another test beam planned for LDMX this year (2025) at ESA at SLAC, only electrons and (possibly) muons will be available there.

Comparisons with Monte Carlo simulations would also be important. They can provide further context to shower shapes beyond the calculated shower maxima and Molière radii. Furthermore, by noting the differences between the data and simulation, the Monte-Carlo simulations could be further fine-tuned, to model instrumental effects more accurately, for example. These tuned models could then be used to produce new and more precise sensitivity plots for LDMX.

As mentioned previously, another test beam will be conducted at End Station A in 2025. The analysis code produced for this thesis will allow for the faster analysis of data taken at the test beam. An updated set of electronics (v3) will also be tested there. Calibration and characterization of these electronics will be done this summer, preceding the test beam.

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